

DEKALB, IL, USA

WMO: 722075

Lat: 41.932N

Lon: 88.708W

Elev: 915

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	0.8	-12.2	2.9	-2.4	-7.3	3.9	2.2	28.6	27.3	26.2	26.3	10.6	280	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	89.2	74.1	86.4	73.2	83.8	72.1	78.2	85.3	76.2	83.1	74.5	81.1	12.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	142.1	83.1	73.8	130.2	80.4	72.3	123.9	79.2	42.4	85.5	40.2	83.1	38.6	81.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	23.2	20.0	-10.6	92.3	6.6	3.1	-15.4	94.6	-19.2	96.5	-22.9	98.2	-27.8	100.5	
			DB												
			WB	-11.2	81.2	6.3	1.8	-15.8	82.5	-19.5	83.6	-23.1	84.6	-27.7	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	22.7	24.4	37.0	48.4	59.6	69.2	72.2	70.1	64.3	51.9	38.8	27.6
	DBStd	19.86	11.95	11.70	11.31	8.96	8.83	6.08	5.68	5.14	7.51	9.07	9.48	10.61
	HDD50	3282	846	719	431	135	17	0	0	0	2	86	351	695
	HDD65	6552	1311	1138	872	500	216	27	8	15	102	419	785	1159
	CDD50	2908	0	1	27	88	314	575	688	622	431	144	16	2
	CDD65	704	0	0	3	3	49	153	231	172	81	12	0	0
	CDH74	5861	0	0	20	51	471	1277	2027	1254	657	104	0	0
	CDH80	1620	0	0	2	2	122	380	626	295	175	18	0	0
Wind	WSAvg	9.7	11.1	11.0	11.0	11.9	10.4	8.8	7.2	6.6	7.6	9.4	10.6	10.6
Precipitation	PrecAvg	34.6	1.9	1.6	2.2	3.1	4.4	4.0	3.7	3.9	2.8	2.8	2.5	1.6
	PrecMax	43.4	3.3	5.2	4.1	5.4	8.0	9.4	7.2	10.1	6.1	6.6	5.4	3.0
	PrecMin	21.7	0.5	0.5	0.9	1.5	0.8	1.0	1.0	0.6	0.4	0.2	0.4	
	PrecStd	5.0	0.8	1.1	1.1	1.3	2.2	2.2	1.7	1.9	1.7	1.8	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	59.4	77.2	79.5	88.7	90.7	94.7	90.3	89.5	82.1	67.9	58.4
		MCWB	50.3	51.9	63.3	61.0	68.3	73.4	78.2	74.5	74.5	69.5	56.9	56.2
	2%	DB	46.1	49.5	65.8	74.2	83.6	88.0	89.5	86.2	85.7	76.9	61.5	50.4
		MCWB	43.8	46.1	56.5	59.7	67.3	72.5	77.2	74.6	72.0	63.6	54.1	47.7
	5%	DB	41.2	42.8	59.4	69.9	79.7	84.6	86.1	83.1	81.2	71.7	57.2	45.5
		MCWB	38.5	39.2	52.1	56.0	64.8	70.8	74.8	72.7	68.5	61.9	52.0	43.2
	10%	DB	37.2	38.0	53.6	64.4	75.3	81.2	83.1	80.8	77.7	67.0	53.3	41.0
		MCWB	35.4	35.6	47.8	54.3	64.0	69.3	73.2	71.0	66.9	60.2	48.4	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	53.4	63.4	65.6	73.8	78.8	81.3	79.4	77.3	72.3	60.7	56.4
		MCDB	53.3	56.0	74.9	73.3	82.9	85.1	88.9	85.5	84.9	79.6	64.4	58.5
	2%	WB	44.2	46.0	58.6	61.7	70.5	75.9	79.1	76.6	74.5	67.9	56.3	47.7
		MCDB	46.1	49.0	65.3	70.0	79.0	83.3	86.6	83.2	81.7	72.5	60.1	49.5
	5%	WB	38.9	40.0	54.4	59.0	68.1	74.0	77.2	74.7	71.9	64.1	52.3	43.0
		MCDB	41.1	42.6	59.2	66.9	76.2	81.6	84.1	80.9	78.3	68.8	56.3	45.0
	10%	WB	35.6	36.0	47.7	55.4	65.6	71.9	75.1	72.8	69.4	60.3	48.5	39.1
		MCDB	37.4	38.1	52.8	64.3	72.6	78.8	81.4	78.4	74.9	65.6	52.9	41.0
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.7	16.0	20.2	19.9	18.9	18.5	18.3	20.1	18.3	15.2	12.3
		MCDBR	15.3	18.3	23.6	27.9	24.6	22.1	20.5	20.6	22.9	23.9	21.4	16.8
		MCWBR	13.2	14.5	16.3	16.0	13.0	11.4	10.9	11.3	11.7	13.8	15.1	14.7
	5% WB	MCDBR	14.7	18.2	21.7	24.2	20.5	19.3	18.8	17.7	19.2	19.9	19.4	15.8
		MCWBR	13.0	15.1	16.2	16.2	12.6	12.0	11.5	11.3	11.2	14.8	15.6	14.5
Clear-Sky Solar Irradiance	taub	0.285	0.303	0.334	0.387	0.413	0.424	0.435	0.420	0.393	0.341	0.300	0.282	
	taud	2.416	2.379	2.403	2.264	2.243	2.254	2.215	2.250	2.303	2.450	2.510	2.450	
	Ebn at Noon	275	287	290	279	274	270	266	266	266	269	268	266	
	Edh at Noon	25	30	33	41	43	43	44	41	36	28	23	22	
All-Sky Solar Radiation	RadAvg	530	826	1151	1475	1698	1928	1928	1695	1415	927	598	440	
	RadStd	62	104	98	121	128	140	107	101	120	108	52	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page